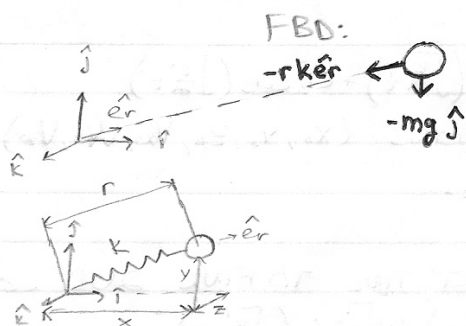


6. MASS HANGING FROM SPRING. 3D. POINT MASS m . ZERO REST LENGTH SPRING, K . GRAVITY g .

a) SET UP EQUATIONS. SET UP FOR NUMERICAL SOLUTIONS. PLOT 2D PROJECTION OF 3D TRAJECTORIES.



$$r \hat{e}_r = x \hat{i} + y \hat{j} + z \hat{k} = \vec{r}$$

BY TRIG,

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\text{SPRING FORCE: } \vec{F}_s = -K r \hat{e}_r$$

$$\vec{F}_s = -K(x \hat{i} + y \hat{j} + z \hat{k})$$

$$\vec{v} = \frac{d}{dt}(\vec{r}) = \dot{x} \hat{i} + \dot{y} \hat{j} + \dot{z} \hat{k}$$

$$\vec{a} = \frac{d}{dt}(\vec{v}) = \ddot{x} \hat{i} + \ddot{y} \hat{j} + \ddot{z} \hat{k}$$

FORCE BALANCE. FIND EOMS.

$\sum \vec{F} = m \vec{a}$ WILL YIELD 3 EQUATIONS FOR 3 UNKNOWN $(\ddot{x}, \ddot{y}, \ddot{z})$

$$-mg \hat{j} - K(x \hat{i} + y \hat{j} + z \hat{k}) = m(\ddot{x} \hat{i} + \ddot{y} \hat{j} + \ddot{z} \hat{k})$$

$$\hat{i}: -Kx = m\ddot{x}$$

$$\hat{j}: -mg - Ky = m\ddot{y}$$

$$\hat{z}: -Kz = m\ddot{z}$$

$$\textcircled{1} \ddot{x} = -\frac{Kx}{m}$$

$$\textcircled{2} \ddot{y} = -\frac{Ky}{m} - g$$

$$\textcircled{3} \ddot{z} = -\frac{Kz}{m}$$

SOLVE W/ ODE45

b) SEE MATLAB CODE

C) JUSTIFY SOLUTION

EXAMINE $\sum \vec{F} = m\vec{a}$

$$m\ddot{\vec{r}} = -k\vec{r} - mg\hat{j}$$

THIS HAS EQUILIBRIUM SOLUTION $\vec{r}_e = -\frac{mg}{k}\hat{j}$

LET $\vec{r}^* = \vec{r} - \vec{r}_e$

→ $m\ddot{\vec{r}}^* = -k\vec{r}^*$ ⇒ THESE ARE 3 HARMONIC OSCILLATIONS OF THE SAME PERIOD. THEREFORE, THIS IS PERIODIC MOTION. THEREFORE, THE PATH WILL ALWAYS BE AN ELLIPSE, AND NEVER WILD OR WIGGLY.

FULL DERIVATION ON NEXT PAGE, IF DESIRED.

DERIVATION

PLAN: SOLVE THE ODE'S. THE SOLUTIONS SHOULD REVEAL SIMPLE HARMONIC MOTION IN THE X, Y, AND Z DIRECTIONS

① $\ddot{x} = -\frac{k}{m}x$ THIS HOMOGENEOUS ODE HAS A SOLUTION OF THE FORM

$$x(t) = C_1 \sin\left(\sqrt{\frac{k}{m}}t\right) + C_2 \cos\left(\sqrt{\frac{k}{m}}t\right)$$

USE INITIAL CONDITIONS $(x_0, y_0, z_0, v_{x0}, v_{y0}, v_{z0})$ TO FIND CONSTANTS C_1 & C_2 .

$$x(0) = C_2 = x_0$$

DIFFERENTIATE WRT TIME TO FIND 2ND CONSTANT

$$\dot{x}(t) = C_1 \sqrt{\frac{k}{m}} \cos\left(\sqrt{\frac{k}{m}}t\right) - x_0 \sqrt{\frac{k}{m}} \sin\left(\sqrt{\frac{k}{m}}t\right)$$

$$\dot{x}(0) = C_1 \sqrt{\frac{k}{m}} = v_{x0}$$

$$C_1 = v_{x0} \sqrt{\frac{m}{k}}$$

$$x(t) = v_{x0} \sqrt{\frac{m}{k}} \sin\left(\sqrt{\frac{k}{m}}t\right) + x_0 \cos\left(\sqrt{\frac{k}{m}}t\right)$$

③ $\ddot{z} = -\frac{k}{m}z$ HAS IDENTICAL FORM: $z(t) = v_{z0} \sqrt{\frac{m}{k}} \sin\left(\sqrt{\frac{k}{m}}t\right) + z_0 \cos\left(\sqrt{\frac{k}{m}}t\right)$

② $\ddot{y} = -\frac{k}{m}y - g$ IS A NON-HOMOGENEOUS ODE, SO IS BROKEN INTO A HOMOGENEOUS AND A PARTICULAR SOLUTION. THE HOMOGENEOUS SOLUTION HAS THE SAME FORM AS $x(t)$ AND $z(t)$

$$y(t) = y_H(t) + y_P(t)$$

CHOOSE A SIMPLE FORM FOR THE PARTICULAR SOLUTION THAT SATISFIES ②.

$$y_P = B$$

$$\rightarrow 0 = -\frac{kB}{m} - g$$

$$B = -\frac{mg}{k}$$

FINALLY, SOLVE FOR UNKNOWN CONSTANTS WITH INITIAL CONDITIONS.

$$y(t) = \underbrace{C_1 \sin\left(\sqrt{\frac{k}{m}}t\right) + C_2 \cos\left(\sqrt{\frac{k}{m}}t\right)}_{y_H(t)} - \underbrace{\frac{mg}{k}}_{y_P}$$

$$y(0) = C_2 - \frac{mg}{k} = y_0$$

$$C_2 = y_0 + \frac{mg}{k}$$

$$\dot{y}(t) = C_1 \sqrt{\frac{k}{m}} \cos\left(\sqrt{\frac{k}{m}}t\right) - C_2 \sqrt{\frac{k}{m}} \sin\left(\sqrt{\frac{k}{m}}t\right)$$

$$\dot{y}(0) = C_1 \sqrt{\frac{k}{m}} = v_{y0}$$

$$C_1 = v_{y0} \sqrt{\frac{m}{k}}$$

$$y(t) = v_{y0} \sqrt{\frac{m}{k}} \sin\left(\sqrt{\frac{k}{m}}t\right) + \left(y_0 + \frac{mg}{k}\right) \cos\left(\sqrt{\frac{k}{m}}t\right)$$

DISCUSSION: $x(t)$, $y(t)$, AND $z(t)$ ARE ALL INSTANCES OF SIMPLE HARMONIC MOTION, SO THE PATH OF THE MASS WILL NEVER BE WILD OR WIGGLY, REGARDLESS OF THE INITIAL CONDITIONS.

Problem 6: Point mass hanging from zero rest length spring in 3D

Contents

- a) Plot 2D projection of 3D trajectories.
- b) Play around with initial conditions to find wildest motion.
- [separate plots](#)

a) Plot 2D projection of 3D trajectories.

b) Play around with initial conditions to find wildest motion.

DISCUSSION: There is no wild motion. The path of the point mass is some sort of ellipse regardless of the initial conditions. The motion is simple harmonic of a constant period in x, y, and z directions.

```
%initial conditions
x0=1; %m
y0=10; %m
z0=-5; %m
vx0=5; %m/s
vy0=-10; %m/s
vz0=1; %m/s
p.g=9.81; %m/s^2
p.m=1; %kg
p.k=1; %N/m

tspan=[0 10]; %integration time (s)

z0=[x0 y0 z0 vx0 vy0 vz0]; %initial state
options = odeset('absTol',1e-9,'relTol',1e-9);
[t,z] = ode45(@(t,z) dynamics6(z,p),tspan,z0,options);
x=z(:,1);
y=z(:,2);

figure(4)
plot(x(1),y(1),'*k')
hold on
plot(x(end),y(end),'^k')
plot(x,y,'k')
axis equal
title('3D Spring Mass. [x0 y0 z0 v_x0 v_y0 v_z0] = [1 10 -5 5 -10 1]')
xlabel('x [m]')
ylabel('y [m]')
legend('start', 'end', 'trajectory')

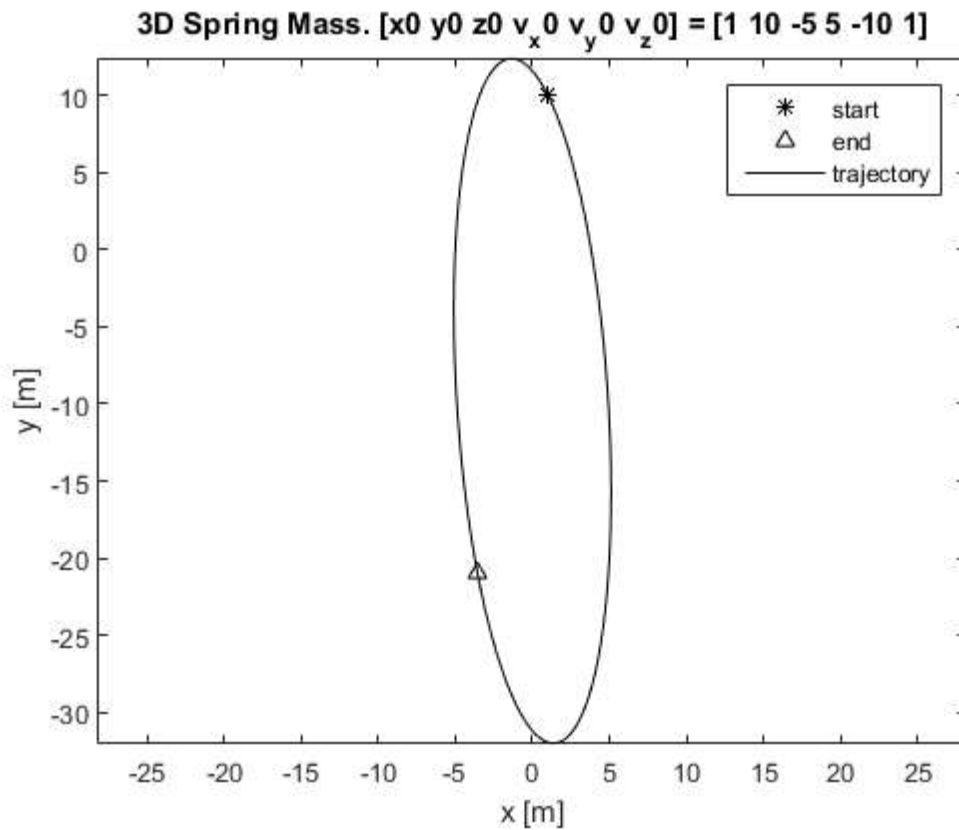
%different initial conditions
x0=-10; %m
y0=-1; %m
z0=1; %m
vx0=3; %m
vy0=7; %m
vz0=-2; %m
p.g=9.81; %m/s^2
p.m=1; %kg
p.k=1; %N/m

tspan=[0 10];
```

```

z0=[x0 y0 z0 vx0 vy0 vz0]; %initial state
options = odeset('absTol',1e-9,'relTol',1e-9);
[t,z] = ode45(@(t,z) dynamics6(z,p),tspan,z0,options);
x=z(:,1);
y=z(:,2);

```

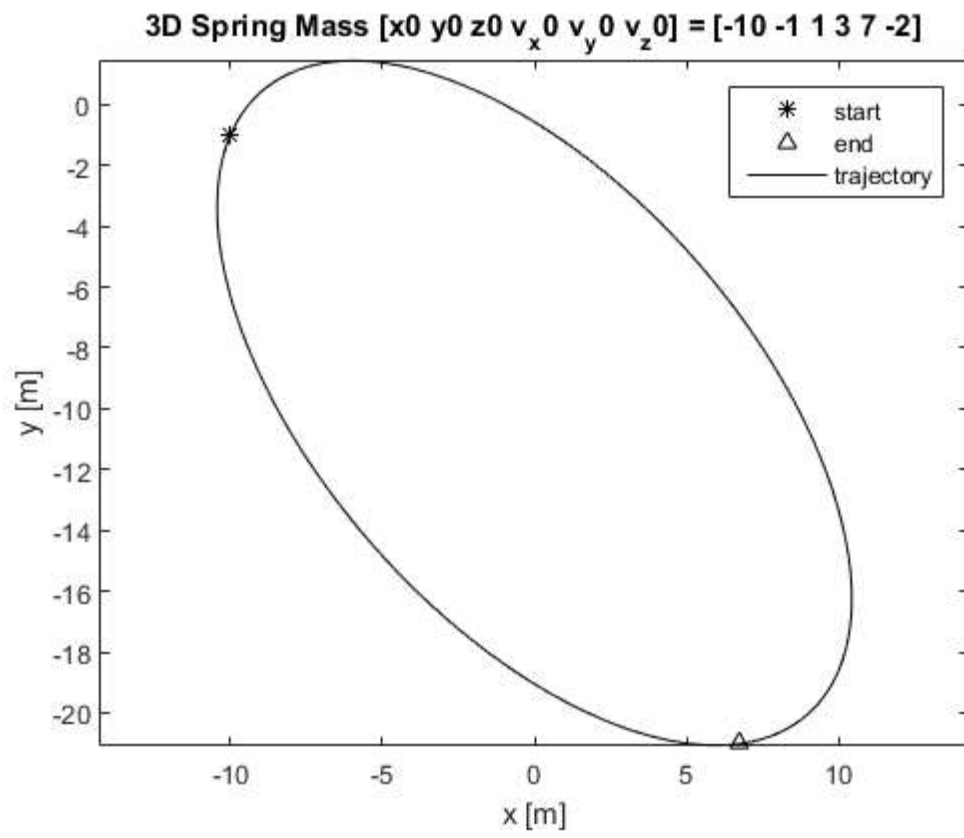


separate plots

```

figure(5)
plot(x(1),y(1),'*k')
hold on
plot(x(end),y(end),'^k')
plot(x,y,'k')
axis equal
title('3D Spring Mass  $[x_0 \ y_0 \ z_0 \ v_{x0} \ v_{y0} \ v_{z0}] = [-10 \ -1 \ 1 \ 3 \ 7 \ -2]$ ')
xlabel('x [m]')
ylabel('y [m]')
legend('start', 'end', 'trajectory')

```



```
function [zdot] = dynamics6(z,p)
% dynamics function, 5730 HW2, problem 6
% z=[x y z vx vy vz]'
% zdot = [vx vy vz ax ay az]'

vx=z(4);
vy=z(5);
vz=z(6);
ax=-p.k*z(1)/p.m;
ay=-p.k*z(2)/p.m-p.g;
az=-p.k*z(3)/p.m;

zdot=[vx vy vz ax ay az]';

end
```